MILITARY SPECIFICATION

RADIO SET CONTROL GROUP, AN/GRA-39( ) (UNITS OF)

This amendment forms a part of Military Specification MIL-R-55154B(CR), dated 24 April 1972, and is approved for use by the Communications-Electronics Command, Department of the Army and is approved for use by all Departments and Agencies of the Department of Defense.

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2.1, DRAWINGS

delete "SIGNAL CORPS" and substitute "CECOM"

add "SC-DL-444445 Connector, Plug Electrical U-229 ( )/U"

"SC-DL-415680 Connector, Assembly U-183/U"

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2.1, DRAWINGS (Cont'd)

delete "SIGNAL CORPS" and substitute "CECOM"

delete "SM-C-456331 Loudspeaker, Permanent Magnet"

3.2, Add the following:

"SC-DL-444445 Connector, Plug Electrical U-229 ( )/U"

"SC-DL-415680 Connector, Assembly U-183/U"

"In addition to the units covered by this specification, Handset H-250( )/U (DL-SM-B-627649) forms a part of the Radio Set Control Group AN/GRA-39 ( )."

FSC-5820

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3.8.7, lines 2 thru 5, delete "except that the . . . SM-C-456331."

3.10, Delete in its entirety and substitute the following:

"3.10 Electromagnetic Interference (EMI). The equipment shall comply with method RE02, Equipment Class CE, Table A-I-MIL-E-55301, Notice 4, MIL-STD-461A."

3.13.2, Delete in its entirety and substitute the following:

"3.13.2 Electrical Preconditioning - Control, Radio Set C-2328 ( )/GRA-39, Contorl, Radio Set C-2329 ( )/GRA-39 or Radio Set Control Group AN/GRA-39( ) (whichever is on order), shall be subjected to a burn-in operation of 48 hours, at 9.0 volts dc input. The last 24 hours of the 48 hours burn-in shall be failure-free. In case of failure, the units must be repaired and the run repeated unitl they successfully complete a 24-hour failure free period. The units shall meet requirements specified herein, without subsequent processing after subjection to 4.7.2."

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4.2c, Delete in its entirety and substitute the following:

"c. Quality conformance inspection before packaging (see 4.5)."

"d. Packaging inspection (see 4.15)."

4.3, TABLE II, Inspection column, delete "Suppression of Radio Interference" and substitute "Electromagnetic Interference."

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4.4, Under ITEM, delete, "Suppression of Radio Interference" and substitute "Electromagnetic Interference."

4.5, Delete "Quality conformance inspection of units before preparation for for delivery." and substitute "Quality conformance inspection before packaging."

4.5, line 3, change "24" to "48."

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TABLE III, Inspection column, delete "Handset H-189( )/U" and substitute "Handset H-250( )/U."

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TABLE III, Inspection Column, Electrical (Cont'd) for Control, Radio Set C-2328/GRA-39 ( ) delete "Microphone circuit frequency Frequency response", "3.8.4" and "4.8.4" and "Keying Relay Circuit Selectivity", "3.8.5" and "4.8.5"

Under Control, Radio Set C-2329/GRA-39 ( ), delete "Tone Generator Output", "3.8.6" and "4.8.6"

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4.5.3, Delete last sentence "Order of test shall conform to Table II."

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TABLE V, Inspection column: delete "Suppression of Radio Interference" and substitute "Electromagnetic Interference".

4.5.3.1.1, lines 1 and 3, delete "three" and substitute "two".

4.5.3.1.3, Delete in its entirety and substitute the following:

"4.5.3.1.3 Sub-Group III - For this subgroup, one unit from the first production lot only will be selected for Group C inspection."

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4.6.4, Delete in its entirety and replace with the following:

"4.6.4 Vibration - Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.7.5.1:

a. Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

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b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement."

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4.7.2, Delete in its entirety and replace with the following:

"4.7.2 Electrical Preconditioning - Control, Radio Set C-2328( )/GRA-39, Control, Radio Set C-2329( )/GRA-39 or Radio Set Control Group AN/GRA-39 ( ) (whichever is on order) shall be continuously operated for a period of 48 hours at a temperature of 40 $\pm$ 5°C (95°F to 113°F) and with an input voltage of 9.0 volts dc. The timer circuit shall be set for a duty cycle of one (1) minute "Transmit", two (2) minutes "Receive", and seven (7) minutes standby. While the units are in this operating mode, vibration shall be performed as follows: 2.2G $\pm$ 10% at any non-resonant frequency between 20 to 60 cycles per second measured at the mounting points of the units. This vibration shall be applied for 10 minutes during each hour of unit operating time. In case of a failure during burn-in, the failure shall be repaired and unit returned until it successfully completes a 24-hour period failure free. As a minimum, the operational test of 4.8.7 herein shall be run to detect failures. Each unit shall be tested for possible failures at least prior to subjection to burn-in and after completion of each 24-hour burn-in/preconditioning period of operation."

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4.8.7, Subparagraph (3), delete "(minimum 2.5 millivolts with normal speaking level)." and substitute "(with normal speaking level)."

4.10, Delete in its entirety and substitute the following:

"4.10 Electromagnetic Interference (EMI). The equipment shall be tested for compliance with 3.10 using the test methods and procedures of Notice 3, MIL-STD-462. The test setup shall be in accordance with Figure REO2-2, Notice 3, MIL-STD-462. The limits shall be in accordance with Figure 13, modified by Figure A-1b, Notice 4, MIL-STD-461A. The simulated two (2) miles of transmission line between units of the equipment shall be accomplished by inserting a 300 ohm non-inductive resistor in each side of the transmission line. When either unit of the control groups is tested independently, the unit not being tested and the two 300 ohm resistors shall be external to the shielded enclosure. During test performance the equipment shall be operated to simulate normal usage."

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4.15, Delete in its entirety and substitute the following:

"4.15 Packaging Inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.15.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.15.2 Quality Conformance Inspection of Packaging.

4.15.2.1 Materials Inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.15.2.2 Preservation Inspection. Inspection of preservation and interior marking shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.15.2.3 Packing Inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VII, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during and identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective."

TABLE VII. PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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4,16, Delete in its entirety.

5, Delete Setion 5 in its entirety (pages 23 through 27) and substitute the following:

"5. PACKAGING"

"5.1 Packaging Requirements. The requirements for packaging shall be in accordance with SPI 1G00059."

TABLE VII, Item Nomenclature column, delete "Handset H-189 ( )/U" and substitute "Handset H-250 ( )/U."

6.2h, i and j, Delete in their entirety and substitute the following:

"h. Level A or B preservation and packing (see Section 5)."

"i. When first article rough handling tests are not required."

"j. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing."

6.3, Delete in its entirety and substitute the following:

"6.3 Color - Final paint finish film color shall be Lusterless Green 383, MIL-C-46168, Film Designation "AG" in Table II of MIL-F-14072."

Add the following new paragraphs:

"NOTES"

"6.10 Level B Preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling."

"6.11 Environmental - Environmental pollution prevention measures are contained in the packaging material specification referenced herein. Refer to material specifications or preparing activity for recommended disposability methods."

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